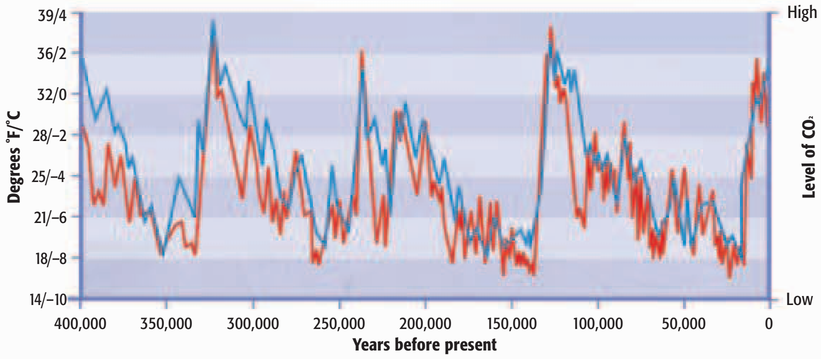
10,000 and 15,000 years ago. Sea cores give only a general impression of Ice Age climate change, but, as a rule, cooling proceeds relatively slowly and warming unfolds rapidly, as was the case at the end of the last cold (glacial) period. Glacial periods in the past have been longer than interglacials—brief, volatile intervals of warmer conditions during the Ice Age when the climate was as warm (or warmer than) today. These increases in temperature are caused by changes in Earth's path around the Sun and its rotation on its axis. Natural increases in greenhouse gases add to the warming. We are currently experiencing an interglacial period caused by these natural phenomena that began about 10,000 years ago.

Environmental change

The Ice Age witnessed dramatic shifts in global climate and major changes in natural environments. During glacial periods, huge ice sheets formed over Scandinavia, and covered most of Canada and parts of the United States as far south as modern Seattle and the Great Lakes. Great glaciers formed on the Alps and there were ice sheets on the Pyrenees, on the Andes, and on Central Asian mountains and high-altitude plateaus. South of the Scandinavian ice sheets, huge expanses of barren landscape extended from the Atlantic to Siberia. These environments suffered nine-month winters and were uninhabitable by ancestors of *Homo sapiens*, who lacked the technology and clothing to adapt to the extremes of temperature. It is no coincidence that *Homo erectus*, with their simple technology and limited cognitive skills, settled in more temperate and tropical environments. The cold caused sea levels to fall dramatically as more water was converted into ice. Huge expanses of what are now continental shelves (land under shallow coastal waters) were exposed, linking land masses—Siberia was part of Alaska, and Britain was joined to Europe. Only short stretches of open water separated mainland Southeast Asia from Australia and New Guinea.

During interglacials, sea levels rose, ice sheets shrank, and forests moved northward as the steppe-tundra vanished. Humans moved north, following the animals they hunted and the plants they foraged, and adapting to a broad range of forested and grassland environments as well as arid and semiarid lands.

90 **The number of meters (yards) sea levels around the world dropped at the beginning of the last Ice Age as water froze to form the ice caps of present-day Antarctica and the Arctic.**



KEY

Level of CO_2

Temperature

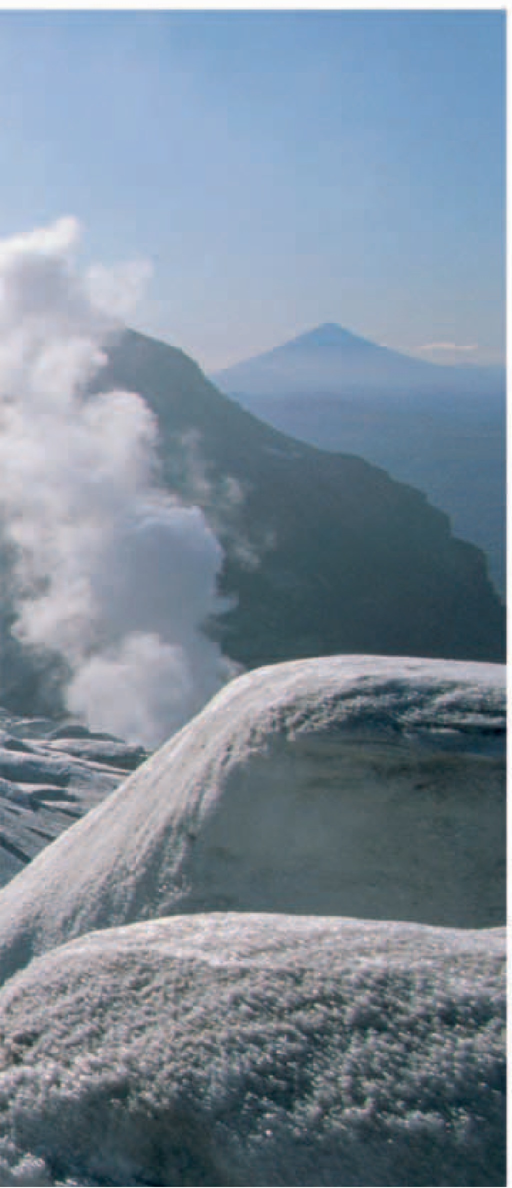
Temperature variations of the Ice Age

Layers of sediment in ice cores taken from Vostok in Antarctica have enabled scientists to chart temperature variations over the past 420,000 years. The levels of CO_2 in the atmosphere have also been recorded and are linked to temperature rises, as can be seen here.

Humans and the elements

The Ice Age climate was volatile and the world's environments changed constantly, which meant that the opportunism and adaptive ability of humans was continuously challenged from one millennium to the next. These challenges may even have been a factor in human evolution. Our earliest ancestors originated in tropical Africa and were basically tropical animals. During long periods of the ice ages, the Sahara was slightly wetter than today. The desert can almost be seen as a pump, drawing in animals and early humans during wetter periods, then pushing them out to the margins when the climate became drier. This was the ecological effect that allowed *Homo erectus* and the animals they preyed on to cross the desert and spread into more temperate environments some 1.8 million years ago.

A major interglacial raised temperatures, peaking around 400,000 years ago. By that time, *Homo erectus* was thriving in north Europe, but they could not adapt to the extreme cold of the glaciation that followed around 350,000 years ago. The few hunting bands living there probably moved southward to more temperate regions. By around 250,000 years ago, there are traces of early human settlement in Europe and parts of East Asia. The final interglacial peaked about 128,000 years ago, when Neanderthals (see p.19) were thriving in Europe. They adapted to the extreme cold of the last glaciation. After 50,000 years ago, modern humans had mastered all the global environments and were living in even the coldest and most extreme parts of the world.



A harsh world

Temperature variations up to 10,000 years ago meant that humans could only survive by adapting to the changing conditions. Our ancestors became successful at surviving and thriving in the cold.

10,000 **The number of years ago that the current interglacial began. Based on past shifts, this warmer phase could last 100,000 years, although the influence of humans may affect this.**

AFTER

Earth is currently experiencing a warmer phase but is still affected by fluctuations in temperature and natural phenomena such as El Niño.

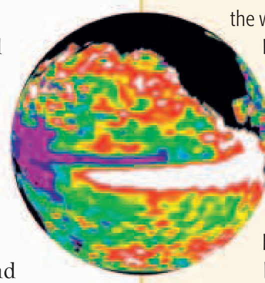
THE ABANDONED SITE OF CHACO CANYON

INCREASED VULNERABILITY

For most of human history, people have lived in small, highly mobile bands **30–31 ››**. Farming **36–37 ››** made humanity more **vulnerable to major climatic events** because people were unable to quickly move to avoid them. Such short-term events were a factor in the **rise and collapse of early civilizations**. One example of this is Chaco Canyon in New Mexico, a site that was settled between 900 and 1150 CE and was abandoned following drought and other unknown dramatic climatic changes.

THE EL NIÑO PHENOMENON

El Niño is a reversal in the flow of water in the Pacific Ocean that causes dramatic changes in the weather every two to seven years. El Niño is one of the most powerful influences on climate after the seasons. The phenomenon originates in the Southwest Pacific and results from interactions between the ocean and the atmosphere. El Niños have affected human history for at least 10,000 years. Major El Niños have **powerful global effects**, causing monsoon failures, and drought or flooding elsewhere. This thermal image highlights El Niño currents in white and red.



EL NIÑO

PERIOD OF STABILITY

As temperatures rose after the Ice Age, humans adapted to a rapidly changing world of **shrinking ice sheets and rising sea levels**. After 5,000 years of irregular warming and cooling, the world entered a warming period that has lasted into modern times. The Vostok ice core tells us this period is among the most warm and stable of the past 420,000 years.

MONSOON SEASON, INDIA

THE FUTURE

The overuse of fossil fuels has increased **global warming**. The future effects of this human-made problem are still unknown.